

CFD on Fuel Manifold

Aryan Yenni

YJSP Bootcamp Spring 2026

Propulsion



Overview

Why Computational Fluid Dynamics?

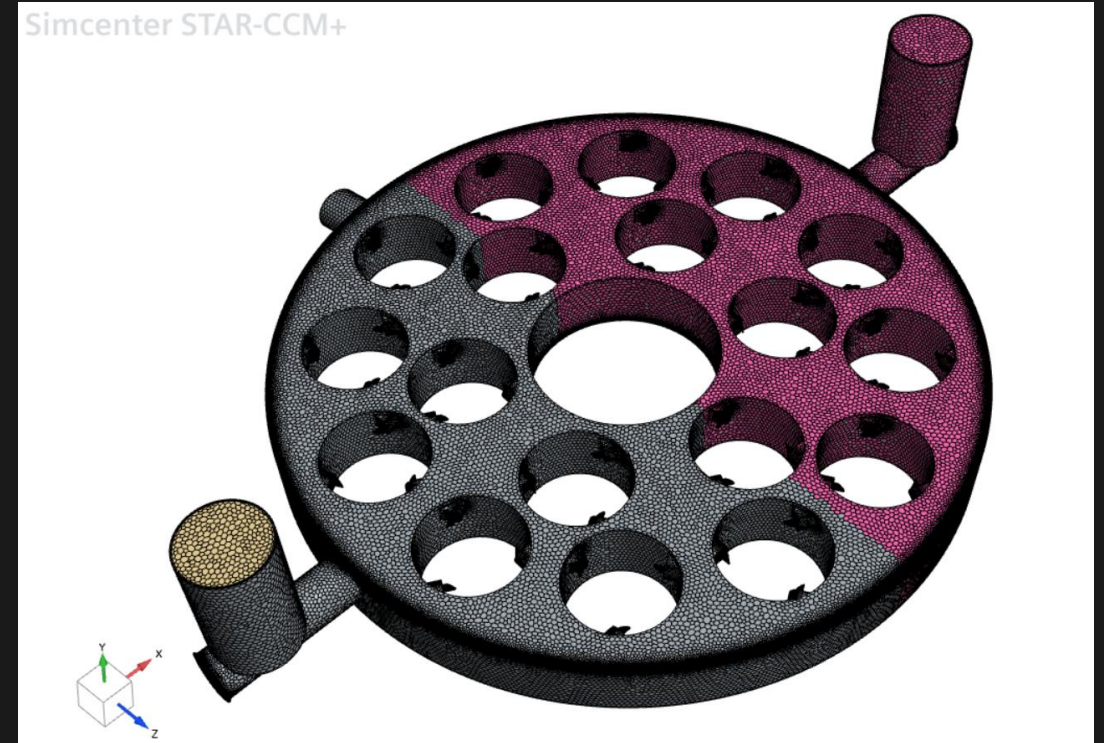
- A tool that simulates fluid motion using math / physics rather than physical experimentation
 - Can find things like velocity fields, pressure gradients, turbulence, mixing, etc.
- Can iterate quickly
 - Change geometry → Run → Compare
- Quite accurate, even with strong assumptions / simplifications
- Saves time, money, effort



Scope

What will we be simulating?

- Incompressible flow within a fuel manifold
- Seeing how the following are affected
 - Pressure
 - Velocity
 - Mixing
 - Mixing ratio
 - Maldistribution
- Last two help us gauge efficiency





Theory

Explaining assumptions based on equations

Incompressible

$$\rho \neq \rho(P)$$
$$\dot{m} = \rho u A$$

Continuum

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$
$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho g \hat{z}$$

Boundary Layers

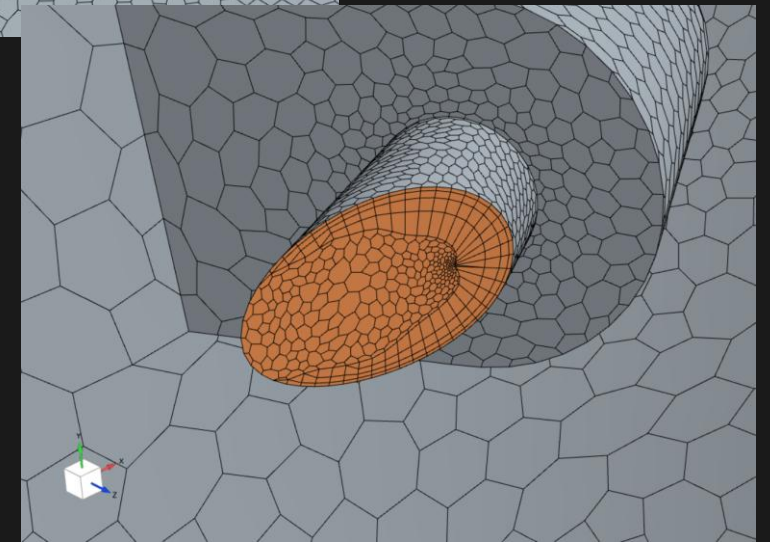
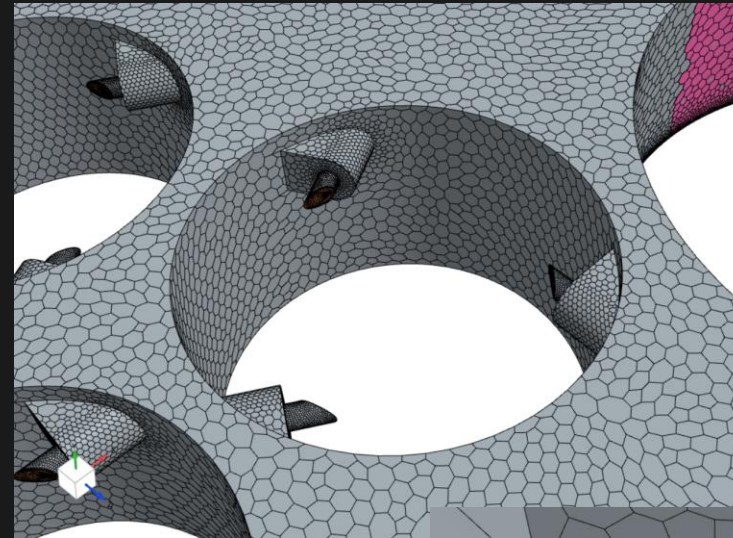
$$u_{wall} \approx 0$$
$$y^+ = \frac{u_\tau y}{\nu}$$



Mesh Scene

What is a mesh?

- Meshes break up the geometry into small areas
 - Easier to resolve values across these small areas
- Strategy
 - Having enough cells across inlets / outlets
 - Prism layers
 - These types of considerations allow solutions to be more accurate

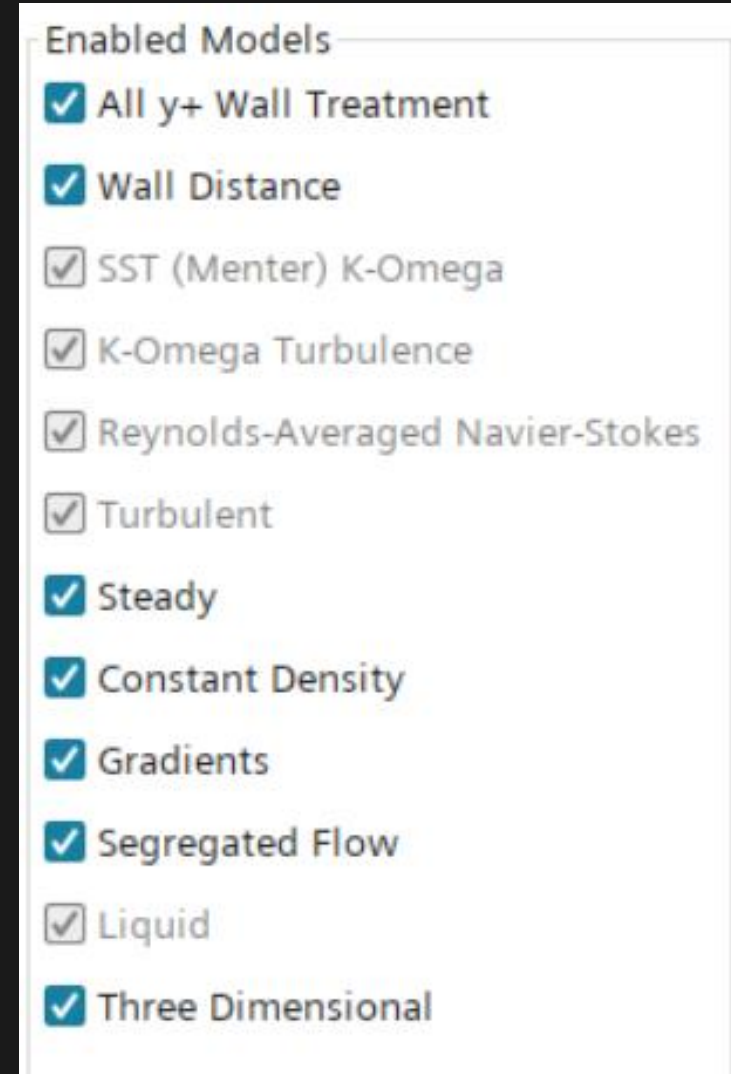




Continua

Telling Star-CCM+ what we're working with

- Covers assumptions
 - State: Liquid
 - Steady state (time independent)
 - Constant density (incomp.)
 - 3-D
 - RANS: $k-\omega$ standard turbulence model
- Makes sure that the software and the engineer are on the same page

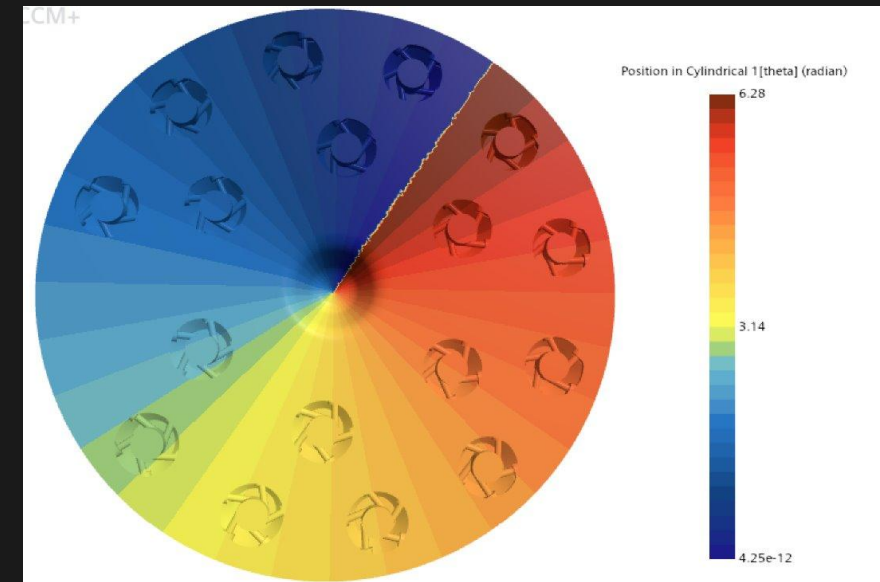
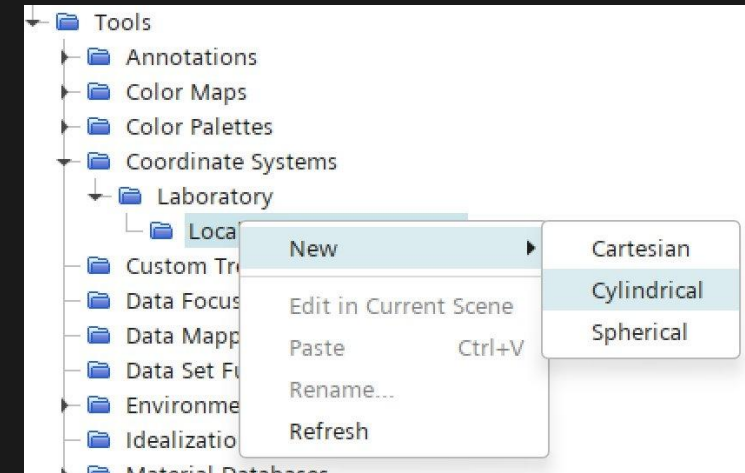




Results

What outputs do we want?

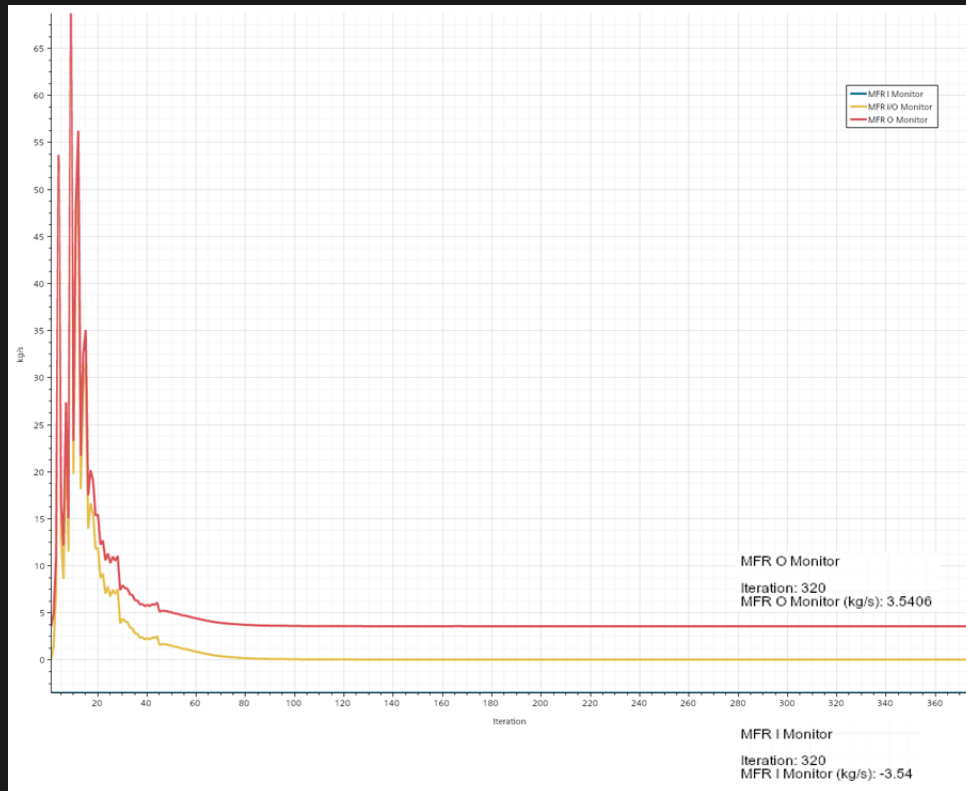
- Reports
 - Mass flow rate
 - Pressure
 - Making sure continuity equations are followed.
- Plotting
 - Mfr vs theta
 - Maldistribution
 - Checking other important parameters because we can.



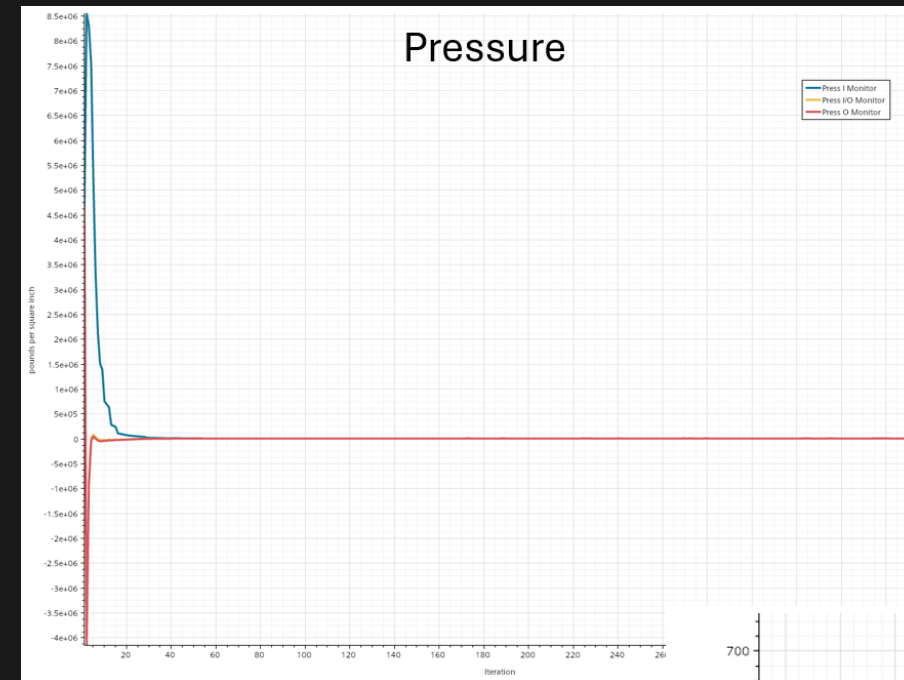


Plots!

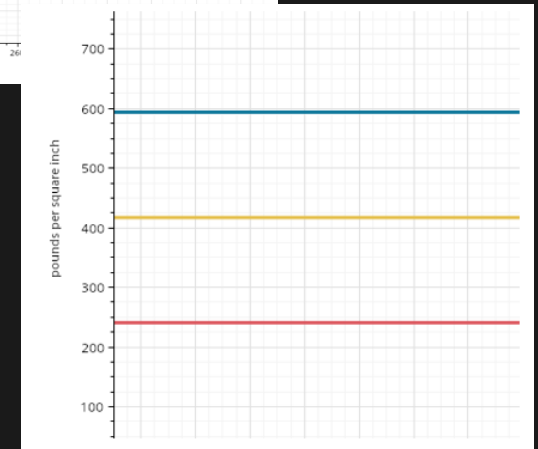
Values converge rather quickly



Mass flow rate



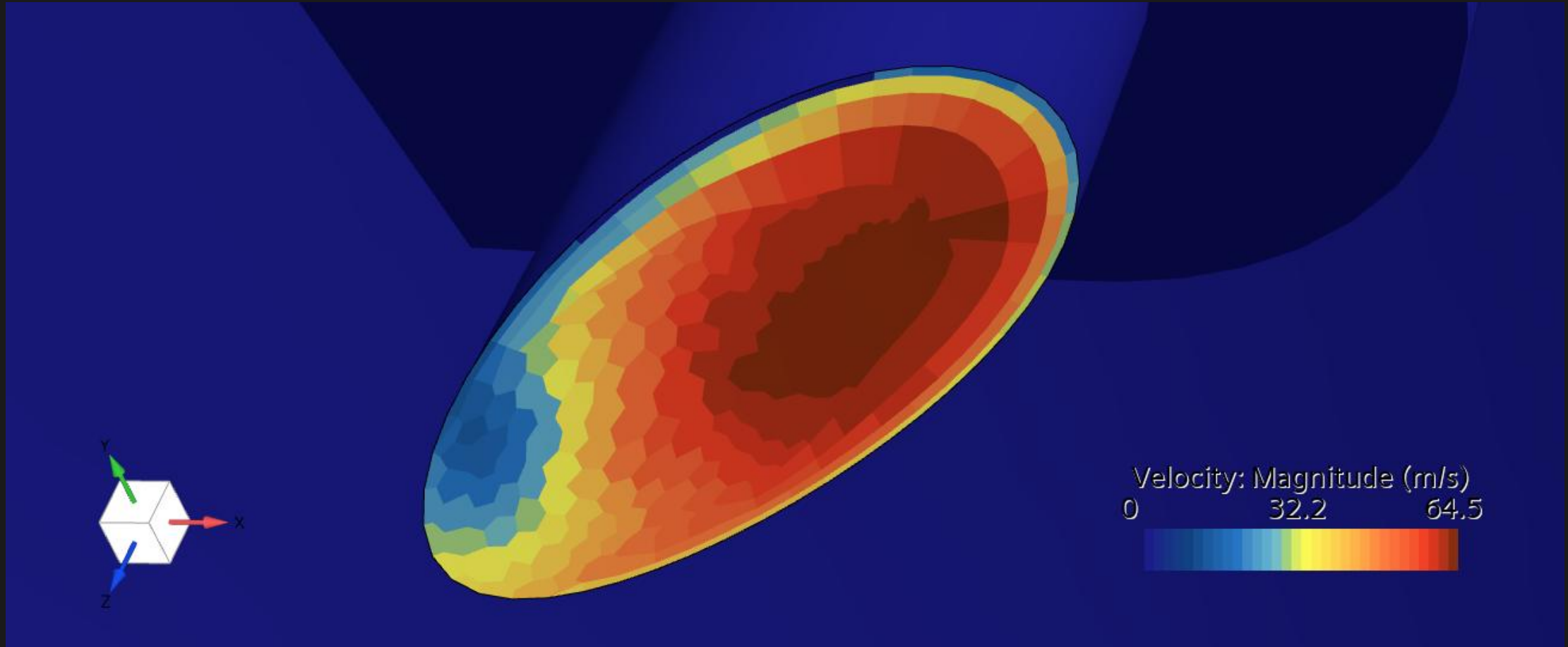
Pressure

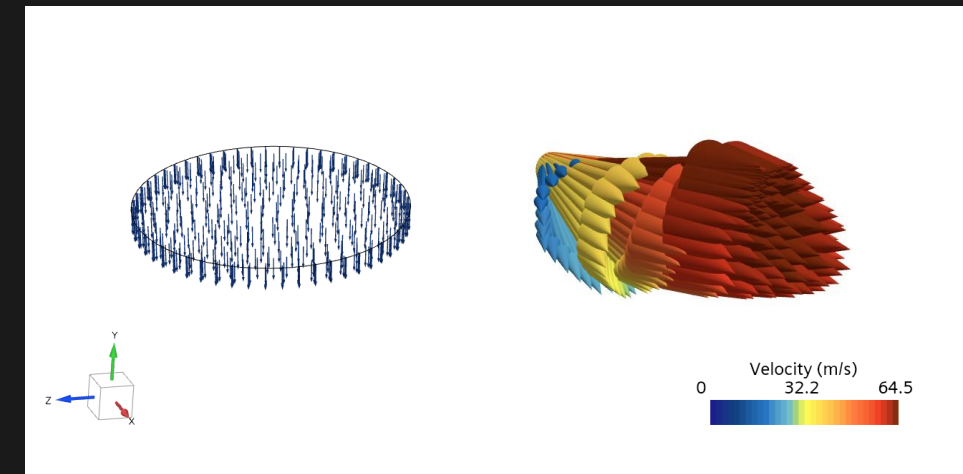
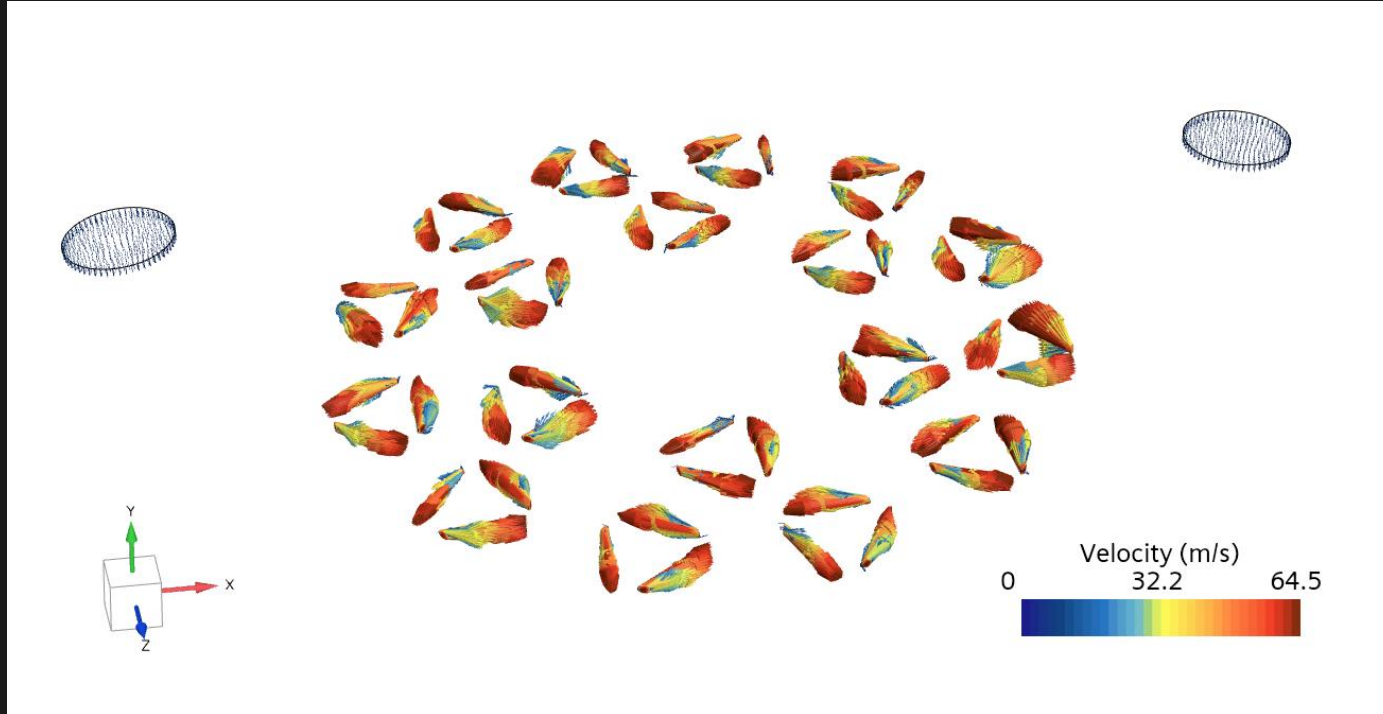




Solution Scenes

Scalar / vector velocity plots, maldistribution







Conclusion

Why this could be important moving forward

- Accurate results before manufacturing and experimentation
 - Reduces risk of design flaws
 - Saves money, time, and effort
- Supports faster iteration
- Establishes a foundation for more advanced modeling in the future!